Lecture plan · for the lecturer

Data quality and project assessment

Data engineering · University · Lecture 4

Time	Materials
60 min	Projector or board; The lecture slides and the seminar sheet from the kit; Phones or laptops with Darsi, for the lab

Learning goals

- Name quality dimensions.
- Test and constrain data.
- Assess real-data projects.

Activity: Clean a real dataset

1. Warm-up (5 min): Show the 112% report and ask students to list possible causes.
2. Check (20 min): Groups write and run three quality checks on the planted dataset.
3. Constrain (15 min): Add constraints that would have refused the bad rows.
4. Defend (15 min): Each group presents and defends one cleaning decision.
5. Wrap-up (5 min): Students write the muddiest point of the lecture and one question; read them before next week.

Discussion

- Which check found the most?
- Which constraint was missing?
- How will you grade the log?

Exit ticket

- Write one quality check for a table you use.

Try it in your next lecture

Hand out a public dataset with planted problems; groups write three quality checks, run them, and each presents and defends one cleaning decision.

Seminar sheet · for students

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Name _____ Group _____ Date _____

1 **Two projects give identical answers. What best distinguishes the stronger one?**

◯ More tables ◯ Longer queries ◯ Quality checks and justified cleaning

2 **Write the check.**

To find duplicate enrolments: SELECT student_id, course FROM enrol GROUP BY student_id, course _____ COUNT(*) > 1; a clean table returns _____ rows.

Word bank: 1 · 0 · HAVING · WHERE

3 **Which are enforced by the database at write time? Choose all that apply.**

☐ FOREIGN KEY (course) REFERENCES course ☐ CHECK (grade BETWEEN 0 AND 100)

☐ A datasheet describing collection ☐ PRIMARY KEY (student_id, course)

4 **Deleting every row with a missing value is always the safest cleaning decision.**

◯ True ◯ False

5 **Match each problem to its quality dimension.**

A student loaded twice	• • • •	Validity
A grade of 140	• • • •	Timeliness
Missing email addresses	• • • •	Uniqueness
Last week’s enrolments not yet loaded	• • • •	Completeness

Answers · for the lecturer

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1 **Quality checks and justified cleaning**

The process shows whether the answers can be trusted.

2 **To find duplicate enrolments: SELECT student_id, course FROM enrol GROUP BY student_id, course HAVING COUNT(*) > 1; a clean table returns 0 rows.**

WHERE filters rows before grouping; HAVING filters groups.

3 **PRIMARY KEY (student_id, course); CHECK (grade BETWEEN 0 AND 100); FOREIGN KEY (course) REFERENCES course**

A datasheet documents; it does not enforce.

4 **False**

It can bias results unless values are missing completely at random; decisions must be justified and recorded.

5 **A student loaded twice — Uniqueness**
A grade of 140 — Validity
Missing email addresses — Completeness
Last week's enrolments not yet loaded — Timeliness

Naming the dimension tells you which test to write.

✎ **For you to think about**

Which real, messy public dataset could your students use next term, and what problems does it contain?